



Larval supply is a limited determinant of settlement at mesoscales across an anthropogenic seascape

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Abstract The relationship between larval supply and settlement is an integral part of the demographic processes of benthic marine organisms that determine their distribution at subsequent life stages. In ascidians, a strong positive relationship between larval supply and settlement has been previously documented, but only at small spatial scales (one location) and short time scales (less than one day). We investigated how this relationship might scale up by sampling larval abundance and settlement of a colonial ascidian (*Botryllus schlosseri*) across mesoscales

(100 s of m) of the anthropogenic seascape of a 1.5-ha marina. Settlement varied vertically with a linear decrease in settlement with increasing depth. This pattern corresponded to the vertical distribution of larvae in the morning, when they were most abundant. However, larval abundance explained only 26% of the variation in settlement at the meso-spatial scale (i.e. the combined effect of the horizontal and vertical dimensions). This weakening of the relationship between larval supply and settlement suggests that the two processes may become decoupled at larger spatial and temporal scales. Our study underscores how changing the scales of sampling may affect our understanding of larval dispersal and settlement processes.

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Introduction

Larval dispersal and settlement are key demographic processes for benthic marine organisms. For benthic invertebrates with biphasic life cycles (i.e. involving the dispersal of larval stages in the water column), recruitment of new individuals to the benthic adult population depends fundamentally on larval supply.

However, until the 1980s, few studies had evaluated the processes and influence of the distribution and abundance of planktonic larvae on the dynamics of adult populations, i.e. supply-side ecology (Connell, 1985; Lewin, 1986; Roughgarden et al., 1987; but see Young, 1987, 1990). Since then, research has been fuelled by a keen interest in determining the role of larval supply on both recruitment rates and subsequent population dynamics of adult stages and benthic assemblages (Underwood & Fairweather, 1989; Pineda et al., 2010). Much of this work has, however, focused on sessile species with planktotrophic larvae, which typically spend weeks in the plankton feeding and dispersing. Barnacles are by far the most well-studied group in this regard and between 75 and 87% of the variation in their recruitment has been explained by larval supply over many different spatial scales, including small spatial scales of < 10 s of m (Miron et al., 1995), mesoscales of 10 s to 100 s of m (Gaines et al., 1985; Minchinton & Scheibling, 1991; Yund et al., 1991), and large scales of 1 s to 1000 s of km (Todd et al., 2006; Broitman et al., 2008).

In contrast to long-distance dispersing species such as barnacles, less is known about the relationship between larval supply and settlement in species with lecithotrophic (i.e. non-feeding) larvae, which spend a considerably shorter time in the plankton and are found in about 10% of marine invertebrate species with pelagic larvae (Thorson, 1950; O'Connor et al., 2007). Regardless, studying this relationship, both under laboratory conditions and *in situ*, is challenging without being able to accurately measure larval supply and settlement and/or manipulate larval behaviour during settlement. However, compared to barnacles and mussels, ascidians are the exception in part because larvae can be cultured without much difficulty for laboratory experimentation (Pineda et al., 2012) and are typically competent to settle within hours of release (Millar, 1971; Boyd et al., 1986; Svane and Young 1989; Marshall et al., 2003). Thus, *in situ* sampling for ascidian tadpole larvae should be a good approximation of larval supply just before settlement.

The supply–settlement relationship in ascidians (especially in brooding species) is expected to be relatively strong because they are typically characterised as closed populations with short-distance dispersing larvae (Olson, 1985; Grosberg, 1987; Davis & Butler, 1989; Stoner, 1990; Bingham & Young, 1991; Petersen & Svane, 1995; Yund & Stires, 2002).

Coupling of larval abundance and settlement should thus occur at small spatial (< 10 s of m) and short temporal (< 1 day) scales. However, large-scale ecological surveys and monitoring activities, such as invasive species surveillance, are often done at substantially larger spatial scales (Sephton et al., 2011; McKenzie et al., 2016; Simkanin et al., 2016). Thus, it is critical to examine how well the supply–settlement relationship can be ‘scaled up’ to meso- (and perhaps larger) spatial and temporal scales to reconcile small-scale observations with large-scale patterns (White & Caselle, 2008; White et al., 2010). Increasing spatial and temporal scales is expected to increase variation in larval supply and settlement due to scale-dependent responses to biotic and abiotic processes (e.g. competition, local hydrodynamics; Menge, 2000; Kinlan & Gaines, 2003; Pineda et al., 2007, 2010). In ascidians, scale-dependent sources of variation in the supply–settlement relationship will affect a relatively narrow range of relevant scales due to their short larval life (typically ranging from several minutes to hours after release; Millar, 1971) and dispersal ability (swimming speeds of 0.2 to 2.5 cm s⁻¹; Svane & Young, 1989). Hence, ascidians are amenable for examinations of larval and settlement processes because contrasts among scales can be made across a smaller range of scales.

Surprisingly, it appears that only two studies (Hurlbut, 1991, 1992) have evaluated the strength of the relationship between ascidian supply and settlement by measuring larval abundance and settlement simultaneously. In these studies, 39% of the variation in larval settlement was explained by larval supply when sampling was done over a narrow range of depths (from 0.25 to 6 m) at a single location over short temporal scales (24 h) and 92% of the variation was explained when bi-hourly sampling was done at a single depth (0.25 m) and location over a 24-h period (Hurlbut, 1991, 1992). Other studies have sampled for both ascidian larval abundance and settlement but separately (Bourque et al., 2007), only larval abundance (Manríquez et al., 2016; Kanamori et al., 2017), or only settlement (Ma et al., 2017; Casso et al., 2018). Further, most of these studies were done only at small (< 10 s of m) spatial scales (but see Manríquez et al., 2016). Thus, it is not known if the strong relationship between larval abundance and settlement at small scales is equally applicable when increasing from the smallest spatial scale (i.e. one sampling location) to

meso-spatial scales of 10 s and 100 s of m and from short (e.g. every two hours) to meso-temporal scales of < 1 week.

As processes affecting larvae also differ between vertical and horizontal dimensions, it is important to explicitly examine larval spatial distributions in three-dimensional space (Pineda et al., 2007). For many species, including crustaceans and fish, larval behaviour and hydrodynamics (e.g. tides) in the vertical and horizontal dimensions interact to drive spatial patterns in larval supply (Hare et al., 1999; Garrison, 1999; DiBacco et al., 2001; Morgan & Fisher, 2010). For ascidians, larval behaviour and settlement are likely to respond more strongly to cues that vary with depth (e.g. light attenuation; Stoner, 1990, 1992; Manríquez & Castilla, 2007), which could decouple larval supply from settlement. In contrast, local hydrodynamic conditions (e.g. currents, winds) are expected to influence horizontal variability in larval supply more than larval behaviour, especially in short-distance dispersing species such as ascidians (Johnson, 1985; Olson, 1985; Koehl, 2007). Consequently, in short-distance dispersing species, a stronger relationship between larval supply and settlement should be expected in the horizontal dimension than in the vertical dimension given the propensity of larval behaviour to decouple the supply–settlement relationship in the vertical dimension.

This study examined the relationship between larval supply and settlement in a closed population of the colonial ascidian *Botryllus schlosseri* (Pallas, 1766) over relevant vertical and horizontal scales, i.e. the scales over which the population is demographically closed. Given the strong supply–settlement relationship observed previously in ascidians and other invertebrate groups, we expected to find a similarly strong (positive) relationship at the meso-scales consisting of multiple sampling locations across a common anthropogenic seascape (a marina). Moreover, we expected that this relationship would be stronger in the horizontal dimension where larval behaviour would have less influence.

Materials and methods

Settlement

Settlement plates were used to document spatial patterns of *B. schlosseri* settlement in Ben Eoin

Marina (45.9818°N, 60.4314°W). The marina basin, approximately 1.5 ha with a maximum depth of 4 m, is located in East Bay of Bras d'Or Lake, Nova Scotia, Canada. Settlement plates (14.6 × 14.6 cm, dark grey polyvinyl chloride [PVC] with one side roughened with a hand-held belt sander) were suspended horizontally in the water column from floating docks at six depths (0.5, 1.0, 1.5, 2.0, 2.5, and 3.0 m below the water surface) with the roughened surface facing downward (Fig. 1). Unlike many studies where plates are 'stacked' in a depth series at each location (e.g. Ma et al., 2017), only one plate was deployed at each location with the treatment *Depth* randomly assigned to ensure spatial independence of sampling locations. Each plate was attached to a brick (four small holes in the plate were used to attach it to a brick with cable ties) suspended at the end of a rope. Plates were deployed on 24 September 2014 during the seasonal period of high settlement rates (K.C.K. Ma, *unpubl. data*). There were 12 replicates for each depth for a total of 72 plates at 72 locations, distributed across the marina basin with a minimum 1-m separation between adjacent plates. Seven days after deployment, the number of *B. schlosseri* recruits on the downward-facing surface of each plate was counted immediately after plate retrieval. Recruits were easily visible without magnification. Although post-settlement mortality in ascidians can be very high (up to 50% after one day and 100% after one month when exposed to predation by grazing fish; Keough & Downes, 1986; Stoner, 1990), recruitment patterns were assumed to reflect initial settlement patterns (e.g. Collin & Johnson, 2014); therefore, the recruitment of *B. schlosseri* recruits integrated over seven days (life stages: oozoids and first blastozoids) on plates is referred to as settlement in this study.

Larval abundance

Plankton samples were collected at half of the plate locations (time constraints prevented sampling at all locations) to document the spatial and temporal variation in larval abundance. 100-L (i.e. 0.1 m³) seawater samples were collected at 1.0-m, 2.0-m, and 3.0-m depths using a 4-m-long hose attached to a diaphragm pump. This seawater sample is likely to represent a relatively localised volume of the water column with no or negligible overlap among targeted depths because depths were spaced a metre apart.

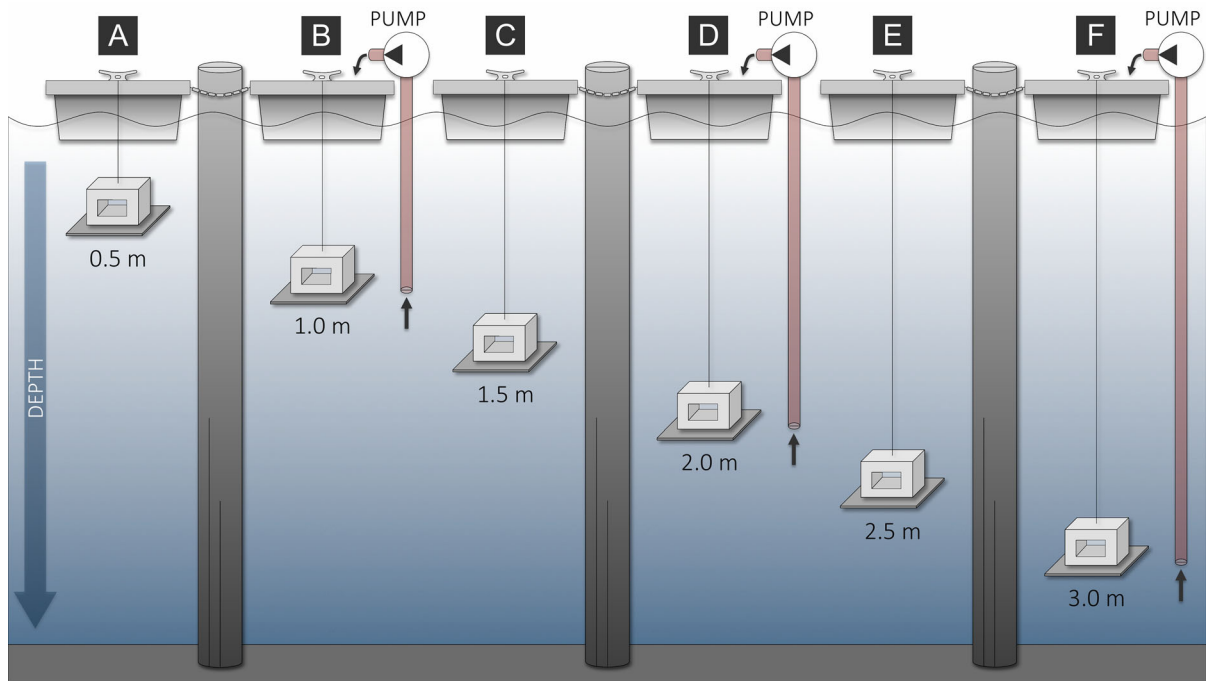


Fig. 1 Diagram of settlement plates suspended from floating dock structures and plankton samples collected via a pump at different depths. **A** Plate at 0.5 m below the water surface;

B plate and plankton sample at 1.0 m; **C** plate at 1.5 m; **D** plate and plankton sample at 2.0 m; **E** plate at 2.5 m; and **F** plate and plankton sample at 3.0 m

Sampling was conducted immediately adjacent (i.e. 20 to 30 cm in the horizontal dimension) to settlement plates at the matching 1.0-m, 2.0-m, and 3.0-m depths so that larval abundance and settlement were sampled at essentially the same location and depth (± 10 cm in the vertical dimension; Fig. 1). Instead of sampling all six depths matching the plates, plankton sampling was limited to these three depths because they are sufficiently spaced apart to represent the targeted depth. After collection, water samples were filtered with a 74- μ m mesh sieve, immediately preserved in 20% ethanol, and *B. schlosseri* larvae counted using a stereomicroscope (Fig. 2). Sampling was done on three consecutive days (28 to 30 September 2014) to account for diel variability, with a total of 36 plankton samples collected each day between 10:30 and 19:15 (i.e. a total of 108 samples). Sampling was concentrated over three straight days and commenced at 10:30 and not any earlier due to logistical constraints. Sampling extended until 19:15 because *B. schlosseri* releases larvae throughout the day (Svane & Young, 1989). However, the order of

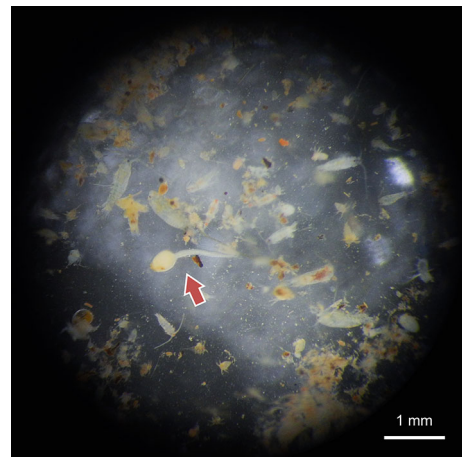


Fig. 2 A *Botryllus schlosseri* larva (red arrow) collected from a plankton sample, preserved in 20% ethanol, and viewed through a stereomicroscope

collecting plankton samples from the 36 locations was randomised each day to minimise potential bias of diurnal variation in larval release or behaviour (i.e. repeated measures of the same 36 locations over time).

Benthic adults

The abundance of *B. schlosseri* was estimated on the vertical sides of floating docks (hard plastic) and the undersides of berthed boats on 27 September 2014 by snorkelling in the upper one metre of the water column. The percent cover of colonies was determined semi-quantitatively (categories: not detected, < 5%, 5 to < 25%, 25 to < 50%, 50 to < 75%, and 75 to 100%) for 20 dock sections (area: approximately 0.5 m²) and 20 boats (length ranging from 6 to 11 m). Due to a lack of artificial surfaces, surveys of benthic adults at deeper depths were not done. Although the marina breakwaters were constructed of large rocks, benthic adults were uncommon on them.

Environmental data

Sensors (INW CT2X) were deployed at each of the six depth treatments to record seawater temperature and salinity values every 15 min on the same days when plankton samples were collected (28 to 30 September). All six sensors were attached to a single line hanging off a floating dock near the geographic centre of the marina basin. Sensors were calibrated prior to deployment. During the study period, water temperature and salinity displayed little variation (mean [range]: 16.3°C [15.6 to 17.0] and 20.2 psu [19.7 to 20.6], respectively). At any given time, differences among depths never exceeded 0.46°C or 0.39 psu.

Data analyses

Variability in settlement

Analysis of variance (ANOVA) was used to evaluate the effects of location within the marina (*Dock*), depth (*Depth*), and *Dock* × *Depth* on settlement to determine horizontal and vertical variability, respectively. A Tukey's honestly significant difference (HSD) test was performed to evaluate variation among depth treatments. To further describe vertical patterns, linear and logarithmic regressions were used to evaluate the relationship between depth and settlement with the expectation that settlement rates in *B. schlosseri* decrease with increasing depth (e.g. Ma et al., 2017).

Variability in larval abundance

A generalised linear mixed model (GLMM procedure with Poisson error distribution and a log link function) was used to evaluate the effects of location within the marina (*Dock*; fixed effect), depth (*Depth*; fixed effect), time of sampling (*Time*; fixed effect; the larval abundance dataset was partitioned post hoc into three intra-diurnal categories: morning, afternoon, and evening), day of sampling (*Date*; random effect), and all two-way and three-way interactions on larval abundance. For a balanced design (i.e. treatments of equal size), the morning time category was defined between 10:30 and 13:10, afternoon was between 13:10 and 16:15, and evening was between 16:15 and 19:15, which resulted in 36 data points per time category. A local regression (LOESS, span = 0.7) curve was applied to evaluate intra-diurnal patterns by showing trends in larval abundance over time of day (data pooled from all three depths and all three sampling days). For a closer examination of vertical patterns, the effect of *Depth* on larval abundance was assessed for each of the partitioned datasets by ANOVA (GLM procedure with Poisson error distribution with a log link function). Tukey's HSD tests were used to evaluate variation among the three depth treatments. For colonial ascidian species, abundance in the water column is expected to peak after larval release earlier in the day (i.e. light-induced larval release) and then to decrease over the course of the day as larvae settle or die (Svane & Young, 1989; Hurlbut, 1992; Forward et al., 2000). Given this expectation, the relationship between time of day and larval abundance was evaluated using linear regressions.

Larval abundance and settlement

The relationship between larval abundance and settlement (data pooled from all three matching depths and all three sampling days) was assessed using linear and non-linear regressions, but an additional non-parametric test (Spearman rank correlation) was used to evaluate the strength of the relationship while taking into account the non-normal distribution of larval abundance data. To ensure that this relationship is true regardless of depth, we examined the assumption of the homogeneity of regression slopes using an ANCOVA procedure on settlement values among depths (factor) while controlling for mean larval

abundances (covariate). Levene's test was used to determine whether this model violated the assumption of homogeneity. To visualise the spatial distribution of deviations within the marina, deviation values (percent deviation of recorded values from those predicted from the pooled regression) were plotted with a colour-contour plot.

Horizontal distributions

Horizontal variations in (i) settlement, (ii) larval abundance (mean over the three days of sampling), and (iii) adults were visualised using colour-contour plot graphs. Because settlement varied with depth (see 'Results' below), settlement data for locations where plates were at 0.5, 1.5, 2.0, 2.5, and 3.0 m were normalised to those of the 1-m depth. Data were normalised by (1) calculating the difference between the mean value for each depth and the mean at 1 m and, then, (2) adding this difference to all the values of this depth. This method preserved depth-specific variance that was observed before and after normalization but not the overall variance because the mean was changed.

ANOVAs (GLM procedures based on type III sums of squares), ANCOVAs, Levene's test, and logistic regressions were computed with IBM SPSS (Version 25). All other statistical analyses were performed in the R programming environment (R Core Team, 2019).

Results

Spatial variability in settlement

Settlement varied significantly as a function of depth and dock (i.e. vertically and horizontally, respectively, within the marina; Table 1). In particular, depth explained about 28% of the variation in settlement (linear regression, $R^2 = 0.28$, $F_{(1,70)} = 18.41$, $P < 0.001$; Supplemental Fig. S1). Over the 2.5-m depth range (from 0.5 to 3.0 m), settlement decreased 35% linearly with increasing depth (Fig. 3). Although the overall relationship is linear (given that the data best fits a linear relationship), settlement was only significantly greater at 0.5 m compared to the four deeper depths (1.5 to 3.0 m; Fig. 3). Equivalent results were found when only three depths were analysed (i.e.

Table 1 Results of an analysis of variance (ANOVA) testing the effects of *Dock*, *Depth*, and *Dock* $\times$ *Depth* on *Botryllus schlosseri* settlement

Source	df	Mean square	<i>F</i>	<i>P</i>
(Intercept)	1	1.329	1869.8	< 0.001
<i>Dock</i>	2	0.012	17.2	< 0.001
<i>Depth</i>	5	0.006	8.2	< 0.001
<i>Dock</i> $\times$ <i>Depth</i>	8	0.002	2.6	0.017

Statistically significant *P* values are bolded

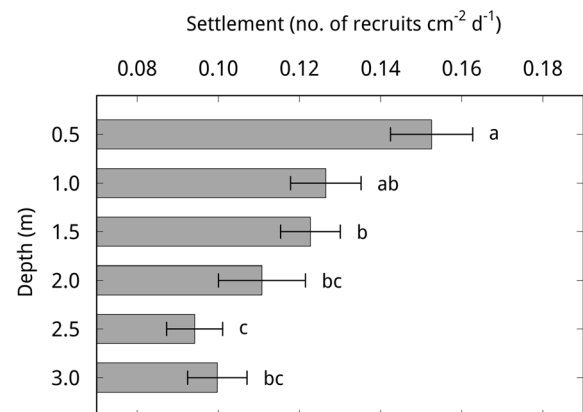


Fig. 3 Vertical distribution of *Botryllus schlosseri* settlement at six depths (bars = standard error; $n = 12$ per depth). Means labelled with the same letter are not significantly different (Tukey's HSD test)

the same depths at which larval abundance were measured), such that there was significant vertical and horizontal variability in settlement (ANOVA GLM; Table 1) and the relationship between settlement and depth was linear (linear regression, $R^2 = 0.12$, $P = 0.04$). Due to the significant interaction between depth and dock (Table 1), the dataset was partitioned both by depth and by dock to further examine spatial variability in settlement with ANOVAs. Overall, significant vertical variability in settlement was observed for each of the three docks in the marina but horizontal variability in settlement was only detected at the three deeper depths (2.0, 2.5, and 3.0 m; data not shown).

Spatial and temporal variability in larval abundance

Although larvae were detected at all times during the day during the sampling period, their intra-diurnal abundance varied significantly (Table 2), such that they were more abundant in the morning (Supplemental Fig. S2). Nevertheless, time of day explained only 15% of the variation in larval abundance (linear regression, $R^2 = 0.15$, $F_{(1,106)} = 19.30$, $P < 0.001$). Also, horizontal variability (i.e. among floating docks) in larval abundance was statistically significant (Table 2). Overall, no statistically significant vertical or day-to-day variation in larval abundance was detected (GLMM; Table 2), although some variation among days can be seen visually (Supplemental Fig. S3). However, larvae were more often detected (i.e. presence vs absence) at shallower depths (81% of the samples taken at 1.0 m, 67% at 2.0 m, and only 56% at 3.0 m). This pattern is supported by a logistic regression, which indicated that depth was a significant predictor for the presence and absence of larvae (Wald statistics = 6.173, $P = 0.013$) although only accounting for 8% of the variation (Nagelkerke $R^2 = 0.08$). GLMM of larval abundance revealed a significant three-way interaction between dock, depth, and time of day but no two-way interactions were significant (Table 2). Moreover, after partitioning into time categories (morning, afternoon, and evening), larval abundance exhibited significant vertical variability in the morning (greater at depths of 1.0 and

2.0 m compared to 3.0 m) but not in the afternoon and evening (Table 3; Fig. 4).

Relationship between larval abundance and settlement

Before proceeding with the regression, we found that settlement did not vary with depth while controlling for larval abundance as a covariate (ANCOVA, $P = 0.338$) and when the assumption of homogeneity of regression slopes was not violated (Levene's test, $P = 0.445$). Larval abundance (mean values of the three sampling days) and settlement were positively correlated when data from all depths were pooled (Spearman rho = 0.52, $n = 36$, $P = 0.001$). However, larval abundance explained only 26% of the variation in settlement (linear regression, $R^2 = 0.26$, $F_{(1,34)} = 12.25$, $P = 0.001$; Fig. 5). The supply–settlement relationship was slightly stronger when larval abundance was $\log_{10}(x + 1)$ transformed (linear regression, $R^2 = 0.30$, $F_{(1,34)} = 9.15$, $P = 0.005$). Further, a logarithmic regression on the untransformed data yielded similar results (logarithmic regression, $R^2 = 0.27$, $F_{(1,34)} = 12.29$, $P = 0.001$). Based on a comparison of AIC scores, the regression model with the highest R^2 value was the best of these three models. When separated by depth, there was only a significant linear relationship between supply and settlement at the 2.0-m depth (linear regression, $R^2 = 0.37$, $F_{(1,10)} = 5.81$, $n = 12$, $P = 0.037$). Visual inspection of residual plots of untransformed data showed random patterns of residuals, which indicate a good fit with a linear model. Notably, the weekly settlement

Table 2 Results of a generalized linear mixed model (GLMM) procedure with Poisson error distribution and a log link function) testing the effects of *Dock* (fixed effect), *Depth* (fixed effect), *Time* (fixed effect), *Date* (random effect), and all two-way and three-way interactions on *Botryllus schlosseri* larval abundance

Source	df	<i>F</i>	<i>P</i>
<i>Dock</i>	2	3.461	0.036
<i>Depth</i>	2	0.601	0.551
<i>Time</i>	2	7.101	0.001
<i>Dock</i> × <i>Depth</i>	3	1.945	0.129
<i>Dock</i> × <i>Time</i>	4	0.126	0.973
<i>Depth</i> × <i>Time</i>	4	1.500	0.209
<i>Dock</i> × <i>Depth</i> × <i>Time</i>	6	2.629	0.022

Statistically significant *P* values are bolded

Table 3 Results of three partitioned analyses of variance (ANOVA, GLM procedure with Poisson error distribution and a log link function) testing the effect of *Depth* on *Botryllus schlosseri* larval abundance in the morning, afternoon, and evening

Partitioned GLM	Source	df	Wald χ^2	<i>P</i>
Morning	(Intercept)	1	46.672	< 0.001
	<i>Depth</i>	2	6.225	0.044
Afternoon	(Intercept)	1	3.997	0.046
	<i>Depth</i>	2	4.685	0.096
Evening	(Intercept)	1	0.124	0.725
	<i>Depth</i>	2	0.391	0.822

Statistically significant *P* values are bolded

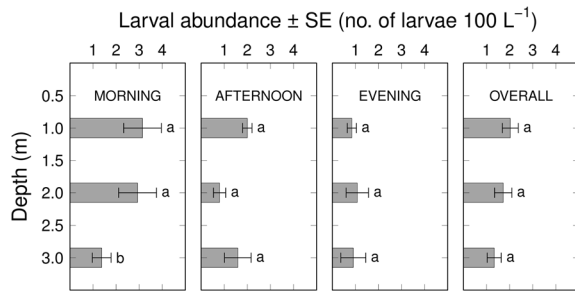


Fig. 4 Vertical distribution of *Botryllus schlosseri* larval abundance at three depth depths in the morning, afternoon, and evening (bars = standard error; $n = 12$ per depth) and for all times of the day combined ('overall'; $n = 36$ per depth). For each panel, the means labelled with the same letter are not significantly different (Tukey's HSD test)

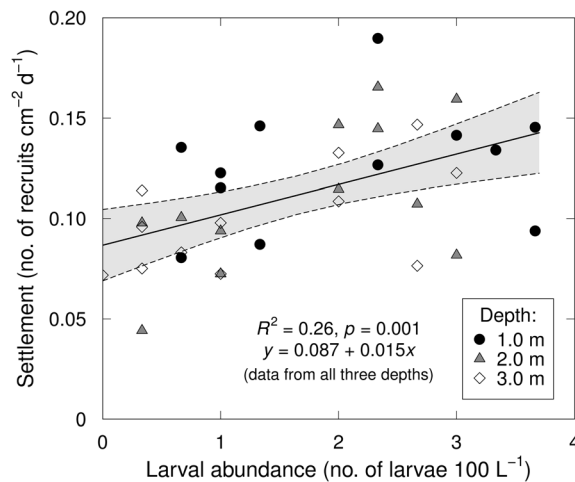


Fig. 5 Relationship between *Botryllus schlosseri* settlement rate over seven days and larval abundance (mean over three consecutive days of sampling); solid line = linear regression (data pooled from all three depths; $n = 36$) and dashed line = 95% confidence limits of the linear regression

on the plates (a mean of 176 recruits plate⁻¹ or 0.83 recruits cm⁻²) corresponded to the number of larvae that can be found in approximately 10 m³ of water (i.e. 0.017 larvae L⁻¹).

Horizontal patterns within the marina

Settlement, larval abundance in the water column (mean over the three days of sampling), and percent cover of adults varied spatially across the marina. Greater settlement (Supplemental Fig. S4a) and larval abundance (Supplemental Fig. S4b) were observed in the western end of the marina basin (around Dock 1)

with an average rate of 0.15 ± 0.02 recruits cm⁻² d⁻¹ ($n = 18$ settlement plates; $\pm$ one standard deviation) and 2.4 ± 2.5 larvae 100 L⁻¹ ($n = 30$ plankton samples), compared to 0.12 ± 0.03 recruits cm⁻² d⁻¹ ($n = 30$) and 1.6 ± 1.7 larvae 100 L⁻¹ ($n = 39$) around Dock 2 and 0.11 ± 0.03 recruits cm⁻² d⁻¹ ($n = 24$) and 1.2 ± 1.9 larvae 100 L⁻¹ ($n = 39$) around Dock 3. However, a greater abundance of adults was observed at the southern portion of Dock 2 and the eastern end of the marina around Dock 3 (Supplemental Fig. S4c).

The spatial distribution of deviations between observed and predicted settlement values (based on the linear regression shown in Fig. 5) varied among docks. The western end of the marina around Dock 1 had an average of 16.1% more settlement than predicted ($n = 10$; Fig. 6) whereas the eastern end (Dock 3) had an average of 13.5% fewer than predicted ($n = 13$). Dock 2, located in the middle, had almost no deviation and, on average, 0.6% more than predicted ($n = 13$). However, variation within each dock was observed, which was most striking on Dock 3 where deviations ranged from 57.4 to -43.0%. Indeed, due to this variation, no significant relationship between larval abundance and settlement at finer spatial scales (i.e. < mesoscales of 100 s of m) was found when the data were partitioned and analysed within floating dock (i.e. Docks 1, 2, and 3).

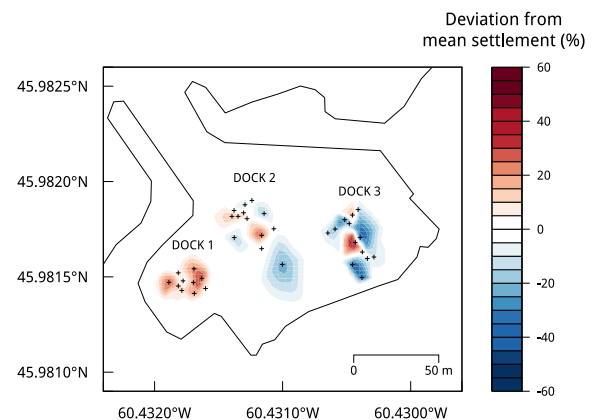


Fig. 6 Percent deviation of observed settlement values of *Botryllus schlosseri* from mean settlement of 0.12 recruits cm⁻² d⁻¹ inside the Ben Eoin Marina, Nova Scotia, Canada. $N = 36$ sampling stations (+)

Discussion

Our general expectation that larval supply should influence settlement rates was observed at mesoscales within the marina with larval supply explaining 26% of the variation in local settlement in this demographically closed ascidian population. Although a positive relationship between larval supply and settlement was indeed observed (because we robustly examined this relationship at the mesoscale), its explanatory power was less than that observed in earlier studies, suggesting that larval supply has a more limited role as a predictor of small-scale and mesoscale settlement patterns than is widely believed or that changes to the spatial scale contributes to substantial variability in these patterns. This result is particularly surprising given that this ascidian species has a short larval period (and, consequently, short dispersal distances, ranging from tens of centimetres to several kilometres) and that the primary source of larvae originates from the adult population of the marina basin (i.e. essentially a closed population; Olson, 1985; Grosberg, 1987; Davis & Butler, 1989; Petersen & Svane, 1995; Yund & Stires, 2002; Collin et al., 2013). Furthermore, this result builds upon our understanding of how larval settlement processes scale up from the small scale to the mesoscale and, more broadly, provides an insight into scale-dependent processes driving ecological patterns (White & Caselle, 2008; White et al., 2010).

In spite of this weak yet compelling relationship, given the day-to-day variability in the horizontal and diurnal distribution of larvae (Supplemental Fig. S3), larval supply appears to play a role in determining settlement patterns with respect to depth. Conceptually, in the vertical dimension, the link between larval supply and settlement can be generalised into four models depending on the combined effects of the vertical distribution of larvae (stratified or non-stratified) and larval behaviour just prior to settlement (selective or non-selective). First, the *non-stratified-non-selective behaviour model* (Fig. 7a) represents the pattern in which larvae inhabit the entire depth range without any vertical stratification in abundance and settle non-selectively on suitable substrata. This model is essentially the null expectation. Second, the *non-stratified-selective behaviour model* (Fig. 7b) represents the pattern in which larvae inhabit the entire depth range without any vertical stratification in abundance but settle selectively on suitable substrata

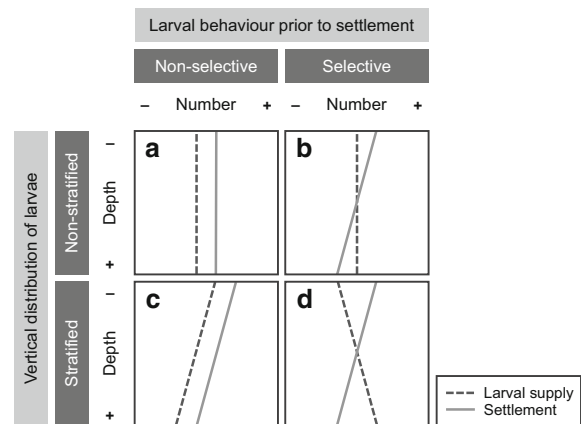


Fig. 7 Conceptualised models of the combined effects of the vertical distribution of larvae (stratified or non-stratified) and larval behaviour prior to settlement (selective or non-selective settlement of competent): **a** *non-stratified-non-selective behaviour*; **b** *non-stratified-selective behaviour*; **c** *stratified-non-selective behaviour*; **d** *stratified-selective behaviour*

at specific depths. Settlement is depicted in Fig. 7b to decrease with increasing depth but alternative linear (e.g. increasing rate of settlement with increasing depth) and non-linear (e.g. a unimodal peak at a specific depth) patterns are also possible scenarios. Third, the *stratified-non-selective behaviour model* (Fig. 7c) represents the pattern in which larvae inhabit specific depths resulting in vertical stratification in abundance and settle non-selectively on suitable substrata. Finally, the *stratified-selective behaviour model* (Fig. 7d) represents the pattern in which larvae inhabit specific depths resulting in vertical stratification in abundance but settle selectively on suitable substrata at specific depths (i.e. either at the same or at other depths). However, one particular scenario of this model, in which vertically stratified larvae selectively settle at the same depths, would result in a pattern that resembles the one depicted in Fig. 7c.

These models can be applied to our data, but they depend on the spatial and temporal scales at which the data are examined. When all data were considered, there was no significant effect of depth on larval abundance. Likewise, when the analysis was limited to settlement data for the three depths where larval was determined (i.e. 1.0, 2.0, and 3.0 m), there were no statistically significant differences in settlement among depths (based on the Tukey's HSD groupings; see Fig. 3). Thus, the pattern would correspond to the one described in the *non-stratified-non-selective*

behaviour model (the null expectation). However, using the settlement data from these three depths, analysis of the data reveals a linear decrease in settlement with depth (which is also the case when all six depths were used in the regression analysis). This pattern corresponds better to the *non-stratified-selective behaviour model*. Finally, if the analysis is restricted to a finer temporal scale, certain situations (i.e. morning) occur when both larval supply and settlement are greater at shallower depths—a pattern corresponding to that of the *stratified-non-selective behaviour model*. This latter model represents the scenario such that one might observe a positive supply–settlement relationship (albeit at fine scales in the vertical dimension). This positive relationship between supply and settlement is an essential condition underlying the conceptual basis of supply-side ecology (Lewin, 1986; Roughgarden et al., 1987). Although seen clearly in other species (intertidal barnacles; Grosberg, 1982), we were only able to observe this association in a limited set of our data. Interestingly, the spatial and temporal scales of sampling can bias how we interpret ecological data and, in this case, select the model that best describes larval settlement. Notwithstanding the sampling design (i.e. larval abundance and settlement were measured at different temporal scales) and caveats (e.g. lack of sampling at night), this study was still able to explain approximately a quarter of the observed variation in settlement and provide evidence of a compelling link between supply and settlement at mesoscales despite all the spatial variation in larval supply and recruitment within the marina and temporal variation in larval supply among days.

Several aspects of this system support the idea that both larval supply and settlement should be higher near the water surface. Larvae should have initially been more abundant in shallower water as most adults were observed in shallow depths on the undersides of floating docks and berthed boats. At very low abundances, deeper populations were restricted to the chains used to anchor the docks or on the deeper rocks used for breakwaters. Outside of the marina basin, *B. schlosseri* colonies were rare and only occasionally found encrusting eelgrass (K.C.K. Ma, *pers. obs.*). Larval behaviour should also contribute to a higher concentration of larvae in shallower depths (Hurlbut, 1991; Stoner, 1992) as ascidian larvae are typically positively phototactic (and, for some species,

negatively geotactic), at least immediately after their release (Grave & Woodbridge, 1924; Crisp & Ghobashy, 1971; Boyd et al., 1986; Svane & Young, 1989). Further, the substantial reduction in boating activities (i.e. nearly no movement of boats within the marina and negligible connectivity with other marinas) during the sampling period in late September and the calm conditions usually found inside the marina (due to the protective breakwaters and the low amplitude of tides in Bras d'Or Lake; Desplanque & Mossman, 2004) likely limited water column mixing, thereby preserving any vertical pattern created by the distribution of adults and larval behaviour (K.C.K. Ma, *pers. obs.*). Finally, there were no indications of abundant ascidian predators (e.g. caprellid amphipods, fish, gastropods, sea stars) that might otherwise prey on larvae, new recruits, or adults (Millar, 1971; Young & Chia, 1984; Olson & McPherson, 1987; Epelbaum et al., 2009; Whitlatch & Osman, 2009; Collin & Johnson, 2014; Rius et al., 2014).

If *B. schlosseri* settlement patterns were driven solely by larval supply, then a variable vertical distribution of larvae (i.e. the *stratified-non-selective behaviour model*) would be the simplest explanation for our observed pattern of greater settlement in shallower depths. However, the resolution of settlement patterns was much greater than that for larval abundance (six depths instead of three), and the decrease in settlement with depth occurred at depths between 0.5 and 2.0 m, where there were no differences in larval abundance (i.e. depth of 1 m vs 2 m). At this finer scale of 0.5-m intervals, larval behaviour may be driving the higher settlement at shallower depths, including the very high settlement at the shallowest depth of 0.5 m. In particular, the general pre-settlement behaviour of ascidian larvae is negatively phototactic, not for swimming to greater depths, but rather to seek dark microhabitats (e.g. crevices, under ledges; Grave & Woodbridge, 1924; Crisp & Ghobashy, 1971; Young & Chia, 1985; but see Forward et al., 2000). Larvae are sensitive to changes in light intensity (e.g. shadows), and abrupt changes may initiate settlement behaviour. For example, a light-to-dark transition has been shown to be a settlement cue for the solitary ascidian, *Ciona intestinalis* (Linnaeus, 1767), as settlement patterns from an *in situ* experiment suggest that larvae swim upwards toward dark surfaces upon experiencing this cue

(Collin, 2013; also see Svane & Young, 1989). Such shade-seeking behaviour should thus be more pronounced at shallower depths where light levels are higher, and thus contrasts are more dramatic when transitioning from light to dark (Olson, 1983; Svane & Young, 1989; Hurlbut, 1993). In this study, our settlement plates were a distinct source of shade for larvae during the daytime. As light attenuation increases with increasing depth, the contrast between light and shade (under the plates) is likely to decrease with increasing depth, in turn, affecting the strength of light as a settlement cue. This mechanism would thus explain the observed linear decrease in settlement with increasing depth, especially at the shallowest depths.

Horizontal variability in ascidian larval supply was observed at the mesoscale (i.e. among docks). While not surprising in of itself, this variation should be linked logically to both the underlying sources of larvae (i.e. the distribution of reproducing adults) and transport after release (i.e. hydrodynamic conditions; Olson, 1985). Given the short ascidian larval period (typically less than a day; Grave & Woodbridge, 1924; Millar, 1971), the lack of pronounced tidal currents (i.e. relatively calm waters; Desplanque & Mossman, 2004), and the semi-enclosed nature of the marina basin (i.e. a closed population), we expected horizontal variation to be driven by the distribution of the adult population. However, despite study limitations such as the lack of plankton sampling earlier at dawn and later at night, this expectation does not appear to be the case as a visual examination reveals the highest adult abundance in the eastern portion of the marina, whereas the highest settlement occurred in the western portion. This spatial pattern could be explained by wind-forced hydrodynamics (e.g. Johnson, 1985). For example, near-shore wind from the west drives the surface waters to transport larvae downwind in the easterly direction. Data from the nearest weather station (Eskasoni, about 13.5 km from the marina) show that the prevailing morning wind was blowing from the west for the first five days then from the east for the remaining two days of the 7-day study period (Environment and Climate Change Canada, 2019). Thus, larvae would be expected to accumulate in the eastern side of the marina, resulting in an uptick in settlement rate assuming that the influence of Ekman transport on larvae is small over these shallow depths (i.e. a small z value). Additionally, at larger spatial and temporal scales, this wind-driven pattern is consistent

with seasonal accumulation and retention of invertebrate larvae in the lee of a headland during the period of upwelling-favourable winds (Wing et al., 1995; Morgan et al., 2011).

Ascidian settlement and subsequent recruitment patterns also have been shown to exhibit spatial variability in the horizontal dimension (e.g. Yund & Stires, 2002; Manríquez et al., 2016). This variation, according to supply-side ecology, should have been driven by larval abundance. However, the link between larval supply and settlement was likewise weak when examined over the larger horizontal dimensions within the marina basin. While these anomalies (both above and below expectations) occurred all across the basin, they were clustered towards the eastern side of the basin. The decoupling of settlement from larval supply may be a consequence of (1) sampled larvae not all being competent to settle, (2) a mismatch in the timing of our larval sampling (a 9-h period of the day) that may have missed any early morning peaks, and (3) the three days of larval sampling did not adequately represent the seven days of settlement activity on plates [see Pineda et al. (2010) for more on the causes of decoupling]. The effects of local hydrodynamics and larval behaviour on dispersal are likely to play a role in the horizontal distribution of competent larvae in the water column (Olson, 1985; Pineda et al., 2010). Although it is not clear why observed settlement rates in certain areas in the marina deviated so greatly from values predicted from larval supply, it is likely that we would have found a stronger relationship between larval supply and settlement if our sampling had been done at smaller spatial and temporal scales as in earlier studies (Hurlbut, 1991, 1992). The observed spatial variation in larval supply and settlement at mesoscales within the marina suggests that certain patterns, such as vertical zonation in larval supply, are more evident at small scales.

In comparison to Hurlbut (1991, 1992), the strong relationship between larval supply and settlement at small scales was reduced and limited at the mesoscale. In our study, the vertical range and number of data points used for the linear regression analysis were similar to Hurlbut (1991), but the horizontal extent was expanded. The increase in spatial extent likely contributed to a loss of predictability as the supply–settlement relationship weakened as the horizontal dimension increased from small scales to mesoscales

of ca. 100 s of m. The weakened relationship may, however, merely be an artefact of differences between different species (*Botryllus schlosseri* vs *Didemnum candidum* Savigny, 1816) and differences between sites (Nova Scotia vs Hawaii); for instance, environmental differences (e.g. temperature) among sites can influence larval development, behaviour, and dispersal ability for a wide variety of marine taxa with an overall tendency of decreasing larval duration with increasing seawater temperature (O'Connor et al., 2007). Likewise, low larval abundance could also contribute to a weakened supply–settlement relationship because the efficacy of sampling methods (e.g. tools, spatial scales of sampling, temporal scales of sampling) may be biased against detecting rare specimens (Pineda et al., 2010). Although we sampled during the seasonal peak of settlement, larval abundance was relatively low compared to other studies [maximal larval abundance can be up to 10 larvae 100 L⁻¹ (Kanamori et al., 2017) or 56 larvae 100 L⁻¹ (Bourque et al., 2007)]. In particular, larval abundance (2 vs 28 larvae 100 L⁻¹) and near-surface settlement rate (0.15 vs 1.06 recruits cm⁻² d⁻¹) were both an order of magnitude lower in our study compared to Hurlbut (1991).

In summary, larval supply and settlement exhibited vertical zonation at mesoscales across an anthropogenic seascape for this closed ascidian population. However, vertical zonation in larval abundance was not maintained throughout the day, and depending on the scale at which the data are viewed, the influence of larval supply on settlement can be understood based on different combinations of vertical distribution patterns of larvae (e.g. stratified and non-stratified) and larval behaviours prior to settlement (e.g. selective and non-selective). Although these combinations of patterns (and the corresponding conceptual models) provide some insight on how larval supply and larval settlement behaviour affect settlement, we conclude that the overall strength of the larval supply–settlement relationship in the vertical dimension was still relatively weak but likely to vary throughout the day, being stronger in the morning and weaker in the afternoon and evening. This relationship was likewise found to be weak in the horizontal dimension, which was inconsistent with our original assumption that a similarly strong supply–settlement relationship could be scaled up to mesoscales. This finding is particularly surprising for such a short-distance dispersing species (< 1 m; Grosberg, 1987) in a semi-enclosed marina

basin (i.e. essentially a closed population) because these conditions should logically promote a strong relationship between supply and settlement. Nevertheless, the supply–settlement relationship is evident given that larval abundance data could explain at least 26% of the variance in settlement determined at a larger temporal scale (seven days), which also demonstrates that the short-term use of settlement plates may be a suitable method, and at best a coarse method, to estimate larval supply (and, further, a better tool to estimate the supply of competent larvae). Finally, determining ecological patterns and processes at mesoscales and describing the general relationship between small scales (the scale at which field experiments are typically done) and mesoscales (the appropriate scale at which ecological phenomena such as larval dispersal occur) are essential to our understanding of scale-dependent processes and patterns in nature.

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Data availability The data generated during the present study are available from the corresponding author upon request.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflicts of interest.

Research involving human and animal rights The authors did not perform any experiments with human participants in this study. Care, handling, and use of animals in this study followed national guidelines.

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